MIL-F-47077(MI)
10 May 1974
SUPERSEDING
MIS-13386A
9 May 1972

MILITARY SPECIFICATION

FABRIC AND TAPES, HEAVY WEIGHT HIGH SILICA, IMPREGNATED WITH A HIGH FIXED CARBON CONTENT, MODIFIED PHENOLIC RESIN

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for extra heavy weight high silica fabric impregnated with modified high fixed carbon content phenolic resin.

1.2 Classification. The heavy weight high silica fabric impregnated with modified high fixed carbon content phenolic resin shall be of the following classes as specified (see 6.2).

a. Class A, broadgoods, fabric, minimum 31 inch width, high pressure laminating.

b. Class B, warp tape slit parallel to the warp direction of the fabric.

c. Class C, bias tape slit at an angle of 45 plus or minus 5 degrees to the warp direction.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on the date of the invitation for bids or request for proposal for a part of the specification to the extent specified herein.

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STANDARDS

Federal

FED-STD-406 Plastics, Methods of Testing

Military

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society For Testing And Materials

ASTM D 1682	Breaking Load and Elongation of Textile Fabrics, Tests for
ASTM D 1777	Measuring Thickness of Textile Materials
ASTM D 1910	Construction Characteristics of Woven Fabrics

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

Uniform Classification Committee

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, 202 Chicago Union Station, Chicago, Illinois 60606.)

American Trucking Association Publication

National Motor Freight Classification Rules and Container Regulations

(Application for copies should be addressed to the National Classification Board, 1424 Sixteenth Street, N.W., Washington, D.C. 20013.)

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Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2) a preproduction sample is required and shall meet the requirements of this specification.

3.2 Material and dimensions. The material and dimensions shall be in accordance with the following.

3.2.1 High silica fabric. The high silica fabric shall be of uniform quality and shall conform to the physical and mechanical properties listed in Table I of this specification.

3.2.1.1 Purity. The silicon dioxide (SiO_2) content of the high silica fabric shall be 98.50 percent, minimum. The sum of sodium oxide (Na_2O), potassium oxide (K_2O) and lithium oxide (Li_2O) shall not exceed 100 parts per million (ppm).

TABLE I

PHYSICAL REQUIREMENTS OF DRY HIGH SILICA FABRIC	
PROPERTY	REQUIREMENTS
Weight-ounces per square yard	38.0 ± 5.0
Thickness (Inches)	0.052 ± 0.008 at an applied pressure of 10 PSI
Areal Shrinkage (percent)	5.0 maximum
Breaking Strength (pounds per inch)	Warp-70 minimum Fill-50 minimum
Thread Count per Inch	Warp-52 ± 4 Fill-40 ± 4
pH	4.5 minimum 7.0 maximum
Width inches (minimum)	31.0
Length in yards Maximum Minimum	 235 35
Type of Weave	12 Harness Satin

3.2.1.2 Additional requirements. Additional requirements such as bagginess, moisture content, and workmanship, may be specified by impregnator, providing there is not a conflict with other requirements of this specification.

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3.2.2 Resin. The resin shall be a phenolic base resin with a fixed carbon content of 56 percent minimum. Quantities of this resin as noted by 5.3.3 shall be supplied with each shipment of fabric or tape for use as a bonding agent between preforms made with the fabric or tape. Resin supplied shall have the same solvents used during the impregnation of the fabric.

3.3 Impregnated fabric and tape requirements.

3.3.1 Dimensions. The minimum width of impregnated fabric shall be 31 inches. The dimensions of Class B and C material shall be as specified by the purchase order, except that tape width tolerance shall be plus or minus 1/16 inch and no more than one roll per width per shipment shall contain less than 40 lineal yards. Unless otherwise stated on the purchase order, no more than one roll per width per shipment shall be other than 17 plus or minus 1 inch outside diameter. The one roll per width per shipment may be less than 16 inches outside diameter.

3.3.2 Handling characteristics. Fabrics shall have sufficient heat tack and drape to allow wrapping on compound surfaces. Tapes shall have the proper tack to permit machine wrapping of oriented configurations.

3.3.3 Tape splices. Rolls of Class B tape shall contain not more than five (5) splices per roll. Class B or Class C tapes shall be joined using sewn butt joint (zig-zag stitch). The thread used shall be nylon or approved equivalent, and shall be kept uncontaminated during any sewing operation.

3.3.4 Storage. All Classes (A, B, and C) of materials when left unopened and sealed in the original containers shall have a minimum life of six (6) months when stored at 10 degrees Celsius (C) ((50 degrees Fahrenheit (F)) maximum.

3.3.5 Physical. All Classes (A, B, and C) of materials shall be made from phenolic impregnated high silica fabric that conforms to the minimum physical and mechanical properties specified in Table II and Table III.

TABLE II

IMPREGNATED FABRIC REQUIREMENTS	
PROPERTY	REQUIREMENT
Volatile Content (percent)	6 maximum
Resin Solids Content (percent)	30 ± 3
Flow (percent)	10 ± 3

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TABLE III

PROPERTIES OF CURED LAMINATES		
PROPERTY	REQUIREMENT (MINIMUM AVERAGE)	
	23.9 ± 5.6 degrees C (75 ± 10 degrees F)	260 ± 5.6 degrees C (500 ± 10 degrees F)
Tensile Strength (psi*) (Warp Direction)	11,000	9,000
Tensile Modulus (psi*) (Warp Direction)	2.2×10^6	1.8×10^6
Tensile Strength (psi*) (Bias Direction)	6,000	5,000
Tensile Modulus (Bias Direction)	1.2×10^6	1.0×10^6
Tensile Elongation (Percent) (Bias Direction)	1.0	1.0
Flexural Strength (psi*) (Warp Direction)	18,000	14,000
Specific Gravity	1.65	--
* psi = pounds per square inch		

3.3.6 Workmanship. Impregnated fabric and tape(s) shall be handled, processed, and stored in accordance with the best practices of the industry. They shall be uniform in quality, clean and free of tears, foreign matter, or other defects detrimental to fabrication or performance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government.

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The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent lots of material. Preproduction samples which do not meet all requirements of this specification shall be subject to rejection.

4.2.1 Preproduction sample size. Samples for preproduction tests shall be provided as follows:

a. Five (5) lineal yards of dry high silica fabric.

b. Forty (40) lineal yards minimum of phenolic impregnated silica fabric a minimum of 4 inches wide to meet Class A requirements.

c. Forty (40) lineal yards minimum of one inch wide (minimum), biased, phenolic impregnated silica fabric tape to meet Class C requirements.

4.3 Classification of inspection. The examination and testing of the material shall be classified as follows:

a. Preproduction inspection.

b. Quality conformance inspection.

4.3.1 Preproduction inspection. Preproduction inspection shall be conducted only on the preproduction samples and shall consist of all the examinations and tests specified herein.

4.3.2 Quality conformance inspection. Quality conformance inspection for acceptance of the impregnated fabrics shall consist of the following examination and tests.

a. Resin solids content.

b. Volatiles.

c. Flow.

d. Dimensional.

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4.3.2.1 Lot size. For purposes of sampling inspection and tests, a lot of dry high silica fabric shall consist of all rolls leached at one time in one bath that are subjected to identical processing after leaching. A lot of preimpregnated fabric shall be defined as all material impregnated from one batch of resin using one lot of dry fabric as defined above.

4.3.2.2 Batch. A batch is defined as that quantity of resin processed through a mill or mixer at one time.

4.3.2.3 Sample size. For each type of material submitted, the supplier shall submit broadgoods samples sufficient to obtain specimens from each sample as required by Figure 1 that are representative of the beginning of the lot, points within the run at intervals of 200 to 235 linear yeards, and the end of the lot. Location of each sample relative to the portion of the run from which it was taken shall be furnished as a part of each sample's identification.

4.3.2.4 Rejection criteria. If the samples of the material fail to meet the quality conformance requirements of 4.3.2 the lot of material represented by the sample shall be rejected.

4.4 Tests methods and examinations.

4.4.1 Dry high silica fabric (unimpregnated).

4.4.1.1 Purity. Emission spectrographic or chemical analysis methods shall be used to ensure that the material meets the requirements of 3.2.1.1.

4.4.1.2 Weight. The material shall meet the weight requirements of Table I and shall be tested in accordance with ASTM D 1910, Section 34 to 40.

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4.4.1.3 Thickness. The material shall meet the thickness requirements of Table I and shall be tested in accordance with ASTM D 1777 using an applied pressure of 10 psi.

4.4.1.4 Areal shrinkage. The material shall meet the areal shrinkage requirements of Table I and shall be tested as follows:

4.4.1.4.1 Sample preparation. Cut at least one 6 inch x 6 inch square specimen from each roll; identify and take at least three (3) measurements each of the length and width to the nearest 0.01 inch. Place the samples in a muffle furnace preheated to 982.2 plus or minus 28 degrees C (1800 plus or minus 50 degrees F) for 30 plus or minus 3 minutes. Remove and cool to room temperature and remeasure as before. In performing calculations, use the average of each dimension before and after firing. Area in both cases is equal to average length times average width.

4.4.1.4.2 Calculation.

$$\text{Percent areal shrinkage} = \frac{(\text{area before firing}) - (\text{area after firing})}{\text{area before firing}} \times 100$$

4.4.1.5 Breaking strength. The material shall meet the breaking strength requirements of Table I and shall be tested as follows:

4.4.1.5.1 Sample preparation. The specimen size for testing of dry high silica fabric shall be 6.0 inches long by 1.25 inches wide with a one inch test width prepared by Method 1R-T of ASTM D 1682. Ends of the fabric shall be covered with masking tape. The jaws of the test grips shall be rubber-faced and are to be separated by three (3) inches at the beginning of the test. The rate of travel of the crosshead shall be 0.5 inches per minute. Five (5) specimens shall be tested in both the warp and the fill direction.

4.4.1.6 Thread count. The material shall meet the thread count requirements of Table I and shall be tested in accordance with ASTM D 1910, Sections 27 to 33.

4.4.1.7 Hydrogen ion concentration (pH). The material shall meet the pH requirements of Table I and shall be tested as follows:

4.4.1.7.1 Sample preparation and testing. Cut a sample of the high silica fabric from each lot of fabric into small pieces and weigh out one gram for testing. Do not handle the sample with bare hands. Place the sample in a pyrex or polyethylene flask (250 milliliter (ml) size) and add 100 plus or minus 1 ml of distilled and freshly boiled

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water. Stir well and heat to 82.2 plus or minus 2.8 degrees C (180 plus or minus 5 degrees F) for 30 plus or minus 1 minutes and allow to cool. Determine the pH of the water mixture using a pH meter. Measure the pH of the distilled water alone for reference purposes. Report both values.

4.4.1.3 Width and length. The material shall meet the width and length requirements of Table I and shall be tested by direct measurement.

4.4.2 Impregnated high silica fabric.

4.4.2.1 Dimensions. The dimensions of Class B and C material shall conform to 3.3.

4.4.2.2 Volatile content. The volatile content shall meet the requirements of Table II. The test shall be run as follows:

a. From each sample, cut three (3) specimens approximately 16 square inches and weigh each to the nearest milligram (0.001 gram) is previously fired at 871.1 plus or minus 28 degrees C (1600 plus or minus 50 degrees F) and tared porcelain crucibles. Record weight (W_1).

b. Dry in a circulating air oven for 15 plus or minus 1 minute at 162.8 plus or minus 5.6 degrees C (325 plus or minus 10 degrees F).

c. Remove, cool to ambient temperature, reweigh to the nearest milligram and record (W_2). If more than one (1) hour is to elapse between removal from the oven and weighing, place the specimens in a desiccator.

d. Calculate volatile content as follows:

$$\text{Percent volatile content} = \frac{W_1 - W_2}{W_1} \times 100$$

Report individual values for each specimen and the average for all three specimens.

4.4.2.3 Resin solids content. The resin solids content shall meet the requirements of Table II and shall be run as follows:

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4.4.2.3.1 Sample preparation. Place the dried specimens from 4.5.2.2 in a muffle furnace at 871.1 plus or minus 28 degrees C (1600 plus or minus 50 degrees F) for a minimum of 30 minutes or until a uniform white color and constant weight is obtained. Remove from furnace, place in a desiccator, cool to ambient temperature, and reweigh to the nearest milligran (W_3). Calculate resin solids content as follows:

4.4.2.3.2 Calculation.

$$\text{Percent resin solids content} = \frac{W_2 - W_3}{W_2} \times 100$$

W_2 is defined in 4.5.2.2. Report individual values for each specimen and the average of all three (3) specimens.

4.4.2.4 Flow. Percent flow shall meet the requirements of Table II. The test shall be run as follows:

a. Cut four (4) 4 inch x 4 inch square specimens on the bias from the impregnated fabric, stack and weigh to the nearest milligram, and record (W_1).

b. Cover the stack, top and bottom, with a glass-reinforced Teflon parting material and place in a press preheated to 162.8 plus or minus 5.6 degrees C (325 plus or minus 10 degrees F) for 20 plus or minus 3 minutes at 150 plus or minus 10 psi. Press closure time shall not exceed five (5) seconds.

c. Remove from press, strip off parting material, scrape off all flash, taking care not to remove any fabric, and reweigh to the nearest milligram (W_2)

d. Calculate resin flow as follows:

$$\text{Percent flow} = \frac{W_1 - W_2}{W_1} \times 100$$

Repeat above test in triplicate and report individual values and the average of all three (3) determinations.

4.4.2.5 Mechanical properties test methods.

4.4.2.5.1 Test specimen preparation. Specimens for tests in Table III shall be prepared from a five ply laminate. The cure cycle shall be 85 plus or minus 5.6 degrees C (185 plus or minus 10 degrees F) at contact pressure for 25 plus or minus 3 minutes, followed by 162.8 plus or minus 5.6 degrees C (325 plus or minus 10 degrees F) at 250 plus or minus 10 psi for 3 plus or minus .17 hours.

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4.4.2.5.2 Tensile properties. Tensile properties shall meet the requirements of Table III and shall be determined in accordance with FED-STD-406, Method 1011. A minimum of five (5) specimens shall be tested for each property. Specimens shall be tested in the warp and bias directions of the fabric.

4.4.2.5.3 Ultimate flexural strength. Flexural strength shall meet the requirements of Table III and shall be determined in accordance with FED-STD-406, Method 1031. A minimum of five (5) specimens shall be tested. Specimens shall be tested in the warp direction of the fabric.

4.4.2.6 Fixed carbon content. The fixed carbon content of the resin shall meet the requirements of 3.2.2. The test shall be conducted as follows:

a. Place three (3) plus or minus 0.5 grams of the resin solution as received from the manufacturer (approximately 60 percent resin solids) into a two-inch diameter aluminum dish for each specimen to be tested. If the resin does not spread to uniformly cover the bottom of the dish, it shall be spread with an appropriate clean spatula to insure a uniform thickness of the specimen.

b. Cure each specimen in a forced air circulating oven following the cure cycle below for oven times and temperatures:

51.7 plus or minus 2.8 degrees C (125 plus or minus 5 degrees F) for 2 plus or minus 0.1 hours followed within 0.5 hours by

65.6 plus or minus 2.8 degrees C (150 plus or minus 5 degrees F) for 4 plus or minus 0.1 hours, followed within 0.5 hours by

93.3 plus or minus 2.8 degrees C (200 plus or minus 5 degrees F) for 4 plus or minus 0.1 hours, followed within 0.5 hours by

121.1 plus or minus 5.6 degrees C (250 plus or minus 10 degrees F) for 4 plus or minus 0.1 hours, followed within 0.5 hours by

148.9 plus or minus 5.6 degrees C (300 plus or minus 10 degrees F) for 4 plus or minus 0.1 hours.

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c. Remove from hot 148.9 degrees C (300 degrees F) oven and place in desiccator until cooled to room temperature. Alternately, the specimen may be allowed to cool to room temperature in the oven and then be conditioned in a desiccator for a minimum of 24 hours before being removed from the dish.

d. Condition a 15 millimeter (mm) to 18 mm diameter by 125 mm to 150 mm long Vycor test tube for 30 minutes minimum in a forced air circulating oven maintained at 176.8 plus or minus 5.6 degrees C (350 plus or minus 10 degrees F). Remove the tube from the hot (176.8 degrees C (350 degrees F)) oven and place in desiccator to cool to room temperature. This test tube shall be left in the desiccator until ready for use. When ready for use, remove the test tube from the desiccator and weigh to the nearest 0.001 gram (W_1).

e. Remove the specimen in its dish from the desiccator and remove the resin from the dish. Carefully break the specimen into small flakes or chunks. Clean lint-free white gloves must be worn during this operation.

f. Weigh out 0.8 to 1.0 gram of the resin chunks or flakes in the preconditioned Vycor test tube. Weigh the resin in the Vycor tube to the nearest 0.001 gram (W_2).

g. Place the Vycor test tube containing the specimen into a clamp mounted onto a ring stand. Connect vacuum hose to glass tube running through a neoprene or Viton stopper. Insert stopper into Vycor test tube to connect the test tube with a vacuum pump, approximately 18 inches of rubber vacuum tubing connected at one end to the vacuum pump and at the other end to a glass T-joint; from the T-joint, one length of rubber vacuum manometer, and another length of rubber tubing approximately 12 inches long shall go to a cold finger trap; from the cold finger trap, a length of the rubber tubing approximately 12 inches long shall go to the glass tubing of the stopper inserted into the Vycor test tube.

h. Evacuate the system described in step g, above, to an absolute pressure of 1.5 mm of mercury or less. Check the system for leaks by clamping off the tubing leading to the vacuum pump for two minutes, minimum. Should the pressure rise by more than 2 mm during the two minutes, recheck the apparatus and seal all fittings with high-vacuum silicone grease.

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i. At constant pressure, place the Fisher burner under the specimen in the Vycor tube, with the flame area completely covering the specimen. The pyrolysis temperature shall be a minimum of 800 degrees C (1472 degrees F) when checked with a pyrometer. The burner shall remain under the sample for 20 to 30 minutes. At the end of this pyrolysis period, the flame from the burner shall be moved so as to drive off as much condensate from the stopper end of the tube as possible without damaging the stopper. At the end of the pyrolysis period the absolute pressure reading shall be less than 1.5 mm of mercury or else the specimen shall be discarded and another run in its place.

j. Shut off the Fisher burner and allow the specimen to cool to room temperature while maintaining an absolute pressure of 1.5 mm of mercury, maximum.

k. Shut off the vacuum, remove the Vycor tube from the vacuum system and place it in a desiccator to further cool for a minimum of two hours, or until constant weight is obtained. Weigh the tube and the pyrolyzed specimen to the nearest 0.001 gram (W_3). The tube shall be handled with clean tools, or if by hand, with the aid of clean, lint-free white gloves only, while being removed from the vacuum system during transfer to or from the desiccator, or during weighing operations.

l. Calculate the fixed carbon content per the following formula:

$$\text{Percent Fixed Carbon Content} = \frac{(W_3 - W_1)}{(W_2 - W_1)} \times 100$$

m. Test five (5) specimens and report the individual values and the average value. The average value shall be used for satisfying the 3.2.2 requirement, except that no single value shall fall below the required minimum value by more than 2.0 percent.

4.5 Shelf life. Six (6) months after the impregnation date, each package of impregnated fabric or tape which has been stored in accordance with the requirements of this specifications shall be retested for resin flow and volatile content in accordance with 4.5.2.4 and 4.5.2.2 respectively. After sampling, the material shall be resealed in accordance with 5.3.1 and 5.3.2. Material meeting the requirements of this specification shall be kept in storage and those which fail on initial or subsequent reinspection shall be clearly identified as being unsatisfactory and shall not be used. This retest procecure shall be repeated every 90 days or prior to use on material past the six (6) month storage requirement.

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4.6 Workmanship. The fabric shall be visually examined for conformance to the requirements of 3.3.6.

4.7 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be examined to determine compliance with Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be as specified herein.

5.2 Preservation. Material shall be shipped under conditions which will prevent deterioration in transit. Storage temperatures of sealed packages of material shall not exceed 10 degrees C (50 degrees F).

5.3 Packaging.

5.3.1 Impregnated fabric (Class A). Fabric shall be rolled on cardboard tubes with a three inch inside diameter. The layers of fabric shall be separated by 0.002 inch minimum thickness polyethylene. In the absence of interleaving material, the fabric layers shall not stick together when shipped and stored as specified in 5.2. The length of the tube shall exceed the width of the fabric, and the roll of fabric shall be end supported by material of sufficient strength to prevent distortion in shipping. Each roll of fabric shall be heat sealed in a polyethylene bag with a minimum wall thickness of 0.002 inch.

5.3.2 Tapes, Class B and C. Tapes shall be wound on individual cardboard tube segments having a length the same as the width of the tape and an inside diameter of three inches. Individual layers of tape may be interleaved with 0.002 inch minimum thickness polyethylene. If interleaving material is not used, the tape layers shall not stick together when shipped and stored per 5.2. Individual rolls shall be heat sealed in a polyethylene bag of 0.002 inch minimum wall thickness.

5.3.3 Container. A one pint container of resin (see 3.2.2) of one of the resin batches used to impregnate the high silica cloth shall accompany each 500 pounds or fraction thereof of Class A, B and C material.

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5.4 Packing. Packaged material shall be shipped in commercial type containers suitably sealed to prevent contamination of the contents by moisture of foreign matter and shall have resealing features. Shipping containers shall be in conformance with the Uniform Freight Classification or National Motor Freight Classification as applicable to the proposed mode of transportation.

5.5 Marking. Shipping containers shall be marked in accordance with MIL-STD-129 and shall include date of packing.

5.5.1 Special markings. Unless otherwise specified (see 6.2), and in addition to the manufacturer's marking, a tag, or label shall be fastened on each carboard roll of fabric and a tag or label placed inside of or on each sealed bag containing fabric, tape, or samples bearing the following information:

a. Purchase order number.

b. This specification number.

c. Class of material.

d. Manufacturer's name and address.

e. Width (fabric and tape).

f. Location of sample from broadgoods lot (Acceptance Samples Only).

g. Lineal yards or weight on roll (fabric only).

h. Net weight of material on roll (tape only).

i. Lot number.

j. Date and place of impregnation.

k. Cloth lot number with manufacturer's resignation.

l. Resin batch number with manufacturer's designation.

m. Storage requirements statement.

PERISHABLE: STORE AT OR BELOW 10 DEGREES C (50 DEGREES F)

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6. NOTES

6.1 Intended use. The material covered by this specification is to be molded or laminated by high pressure into ablative and insulation composites.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Applicable levels of preservation, packaging and packing (see 5.1).

d. Sampling plan if other than specified (see 4.3.1 and 4.3.2).

e. Class and cut required (see 1.2 and 3.3.1).

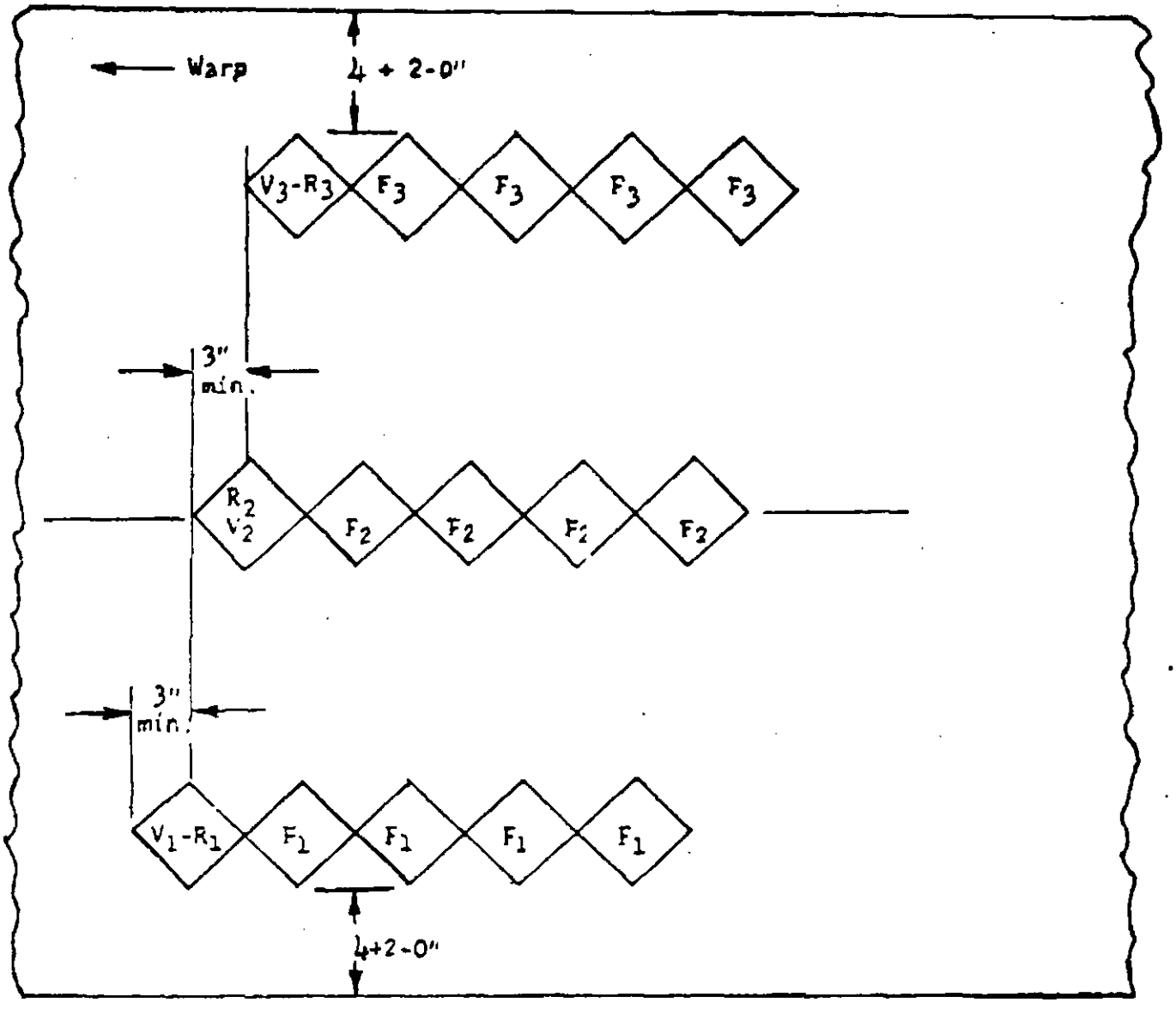
f. Width of Class B and C (Tape) material.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-13386A, dated 9 May 1972.

Custodian:
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Preparing Activity:
ARMY - MI
Project No. 8305-A316

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V - R = Volatile and Resin Solids Content Specimens.

F = Flow Specimens

Figure 1

Specimen sampling plan for each broadgood sample

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